

Adaptive Training Services for Energy-Constrained Satellite Constellations

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Abstract. We propose a novel federated learning framework that leverages run-time adaptive mixed-precision strategies to sustain AI model training under dynamic energy constraints in satellite constellations. The framework models local model training as an adaptive AI service with a key additional constraint - the satellite’s battery budget. Its functional objective is to generate local model updates while dynamically adapting non-functional attributes, including numerical precision, computational cost, latency, and energy consumption. To navigate this design space, we introduce two additional controls. The first is to continuously employ adaptive precision quantization, dynamically switching between FP32, BM16, and BM8 (8-bit block-mantissa) representations at run-time to balance energy consumption and model accuracy during training. The second is a hierarchical battery-aware backpropagation mechanism that selectively updates different portions of the network based on battery state. We validate the proposed training framework on the CIFAR-10, MNIST, and EuroSAT datasets. Results show that a combined mixed-precision and backpropagation depth control mechanism preserves model convergence characteristics comparable to full-precision training when some training steps are skipped under constrained energy conditions.

Keywords: Training Service · Satellite Service · Service Composition · ResNet18
· CNN · Neural Networks

1 Introduction

Rapid deployments of large-scale satellite constellations are transforming space-based computing by enabling distributed AI and machine learning services directly onboard satellites. In this regard, federated learning provides a service composition mechanism for collaboratively training machine learning models

across geographically dispersed satellites without transferring raw data to ground stations [5]. However, unlike terrestrial computing environments, satellite services operate under highly dynamic and often unpredictable energy conditions [16]. Available power can vary significantly due to orbital position, eclipse periods, solar panel orientation, battery degradation, and mission-specific service workloads [17]. These conditions result in substantial fluctuations in the power available to individual nodes, meaning energy-efficiency is a critical requirement for an AI training service in space-based federated learning environments [16].

A key challenge in this domain is to adapt the training service to varying energy availability while still achieving acceptable model performance. Conventional neural network training services commonly rely on full-precision computation, such as Floating Point 32-bit (FP32) arithmetic, together with complete backpropagation through the network. As battery levels decrease, satellites may no longer have sufficient energy to provide conventional full-precision training as a continuous onboard service. To save energy in such situations, existing techniques include continuing training with reduced computational requirements or suspending training service altogether [18]. To reduce computational load, one method is to reduce numerical precision during training using formats such as Block Mini Float 16-bit (BM16), Block Mini Float 8-bit (BM8), or other low-precision representations [3]. However, the problem with this approach is that while lower precision reduces energy consumption, it also impacts accuracy; so the challenge is to ensure such a reduced-cost training service can still provide a meaningful contribution to the global federated model under constrained battery conditions. Alternatively, training costs can be reduced by limiting how far gradients are propagated through the network during backpropagation. Instead of updating all layers, partial or hierarchical backpropagation updates only selected layers, such as later convolutional blocks or the classification head; this reduces both computation and memory overhead [22], but it again comes at a cost of accuracy. Identifying the optimal combination of precision levels and backpropagation strategies under battery conditions that dynamically change at run-time remains an open research problem.

Existing work approaches the problem of energy-constrained distributed training in several directions. In the satellite federated learning setting, Razmi et al. propose an energy-aware computation time scheduler for LEO constellations that minimizes battery usage by pacing training workloads relative to solar harvesting cycles and eclipse periods [20]. The results demonstrate more than a three-fold increase in battery lifetime over energy-agnostic scheduling. Infrastructure-level approaches include FedLEO [31], which augments conventional federated learning’s star topology with intra-plane communication pathways. By exploiting the predictability of orbital patterns, FedLEO enables model propagation and optimized scheduling between sink satellites and the ground station. At the compute level, mixed-precision training has emerged as the dominant technique for reducing per-operation energy cost [6]. It uses lower-precision formats such as FP16 or BM16 for most operations to improve speed, while retaining critical parts in FP32 to preserve numerical stability. More aggressive formats such as FP8

can further reduce activation storage and increase throughput, but may introduce training instability [13]. On resource-constrained devices, backpropagation is both energy- and memory-intensive due to limited on-chip memory, motivating selective fine-tuning approaches such as MiniLearn [19], which fine-tunes dequantized hidden layers while retaining the remainder of the network in quantized integer precision, reducing energy usage without compromising performance.

Despite these advances, the existing literature treats precision reduction, partial backpropagation, and energy scheduling largely in isolation; no prior work systematically characterizes the joint optimization of numerical precision and backpropagation depth as a function of battery state. Instead, these federated learning frameworks typically assume that each participating node can provide a stable training service with sufficient power and computational resources throughout execution [2]. Moreover, studies exploring low-precision training and energy-efficient machine learning consider the training service using a fixed precision configuration that remains unchanged at runtime [27]. This restriction is not ideal for the case of service provided by satellite constellations, where energy availability can vary significantly between satellites and throughout an orbit [30]; using more energy when available is more desirable to increase the effectiveness of the federated learning process.

In this paper, we treat LEO satellites as distributed edge service providers capable of onboard analytics, local learning, and federated collaboration [11]. We consider a clustered satellite constellation in which multiple worker satellites are coordinated by a master satellite. Within this setup, each worker satellite dynamically adapts its local training service by adjusting numerical precision and backpropagation depth according to its available energy budget, while the master satellite coordinates the collection and aggregation of model updates at the cluster level [28]. We introduce run-time adaptable mixed-precision (RAMP) training and hierarchical back propagation as service control mechanisms. RAMP training uses different numerical formats within the same training process to reduce computation, memory usage, and energy consumption while maintaining acceptable learning performance. Depth control regulates how much of a CNN is updated during backpropagation by selectively training last few layers. Using these key service controls, we propose an energy-aware federated learning framework that delivers onboard AI training as an adaptive service. Each satellite provides a local training service whose functional objective is to contribute model updates to the federated learning process, while its non-functional attributes—such as energy consumption, computational cost, precision level, latency, and training quality—are dynamically adjusted according to available energy resources. In particular, RAMP and hierarchical backpropagation act as service-level control mechanisms, allowing each satellite to reduce training cost while preserving useful participation in the federated-learning-based service composition. We evaluate the proposed approach on CIFAR-10, MNIST, and EuroSAT datasets, with an HLS-based implementation designed according to the resource capabilities of the Xilinx ZCU102 FPGA platform. The results demonstrate that adaptive precision and depth-controlled training can maintain model convergence while

reducing computational intensity under constrained energy conditions. The main contributions of this paper are as follows:

- We propose an energy-aware federated learning framework for satellite constellations that models local AI training as an adaptive onboard service. It enables energy-constrained nodes to continue participating in the training.
- We introduce an adaptive training service, running on an FPGA, which supports block-mantissa precision with hierarchical backpropagation, reducing training cost through dynamic precision and update-depth control.
- To support reproducibility and independent validation, we make our code publicly available⁴.

This paper focuses specifically on run-time adaptive local training services under dynamic energy constraints and does not investigate aggregation strategies within the federated learning process. In particular, the proposed framework does not optimize how updates from satellites operating at different precision levels or backpropagation depths should be combined at the global model level. Instead, aggregation is treated as a standard federated learning operation, while the core contribution of this work lies in enabling energy-constrained satellites to continue local training through adaptive precision and hierarchical backpropagation.

Motivating Scenario In satellites, eclipse periods are a recurring source of power stress because satellites temporarily lose solar energy generation and must depend on stored battery capacity. When onboard workloads are energy-intensive, satellites may be unable to sustain normal service delivery. A real-world example of eclipse-induced operational limitation can be observed in the AO-85/Fox-1A CubeSat, which was turned off after eclipse periods caused precarious battery-voltage drops under degraded battery conditions [1]. The satellite had previously experienced precarious battery-voltage drops during eclipse, and operators suspected that its batteries had degraded due to thermal stress during earlier periods of continuous illumination. As AO-85 re-entered regular eclipse cycles, maintaining normal operation risked further battery damage. To preserve the satellite’s usable life, operators disabled both the internal housekeeping unit and the transmitter, which stopped all transmissions, including periodic beacon transmissions normally sent during sunlight and eclipse. The satellite was later enabled during a full-sun illumination period, but it continued to experience low-voltage resets. This case demonstrates that eclipse-driven power shortages can force satellites to suspend computational and communication workloads. In the context of federated learning, similar power limitations could prevent a satellite from completing local training or contributing timely updates. As illustrated in Fig. 1, the motivating scenario contrasts two possible behaviors for satellites operating under different battery conditions. Without adaptation, all satellites are expected to execute the same full-precision training workload, which allows high-battery satellites to complete training but causes medium- and low-battery

⁴ <https://github.com/MuhammadUmairMujahid18/RAMP>

satellites to miss updates or drop out when energy becomes constrained. In contrast, the energy-aware adaptation case allows each satellite to adjust its training service according to its available energy: high-battery satellites perform full training, while energy-constrained satellites switch to reduced precision and partial backpropagation. This graceful service degradation enables satellites to continue contributing useful local updates during power-constrained periods.

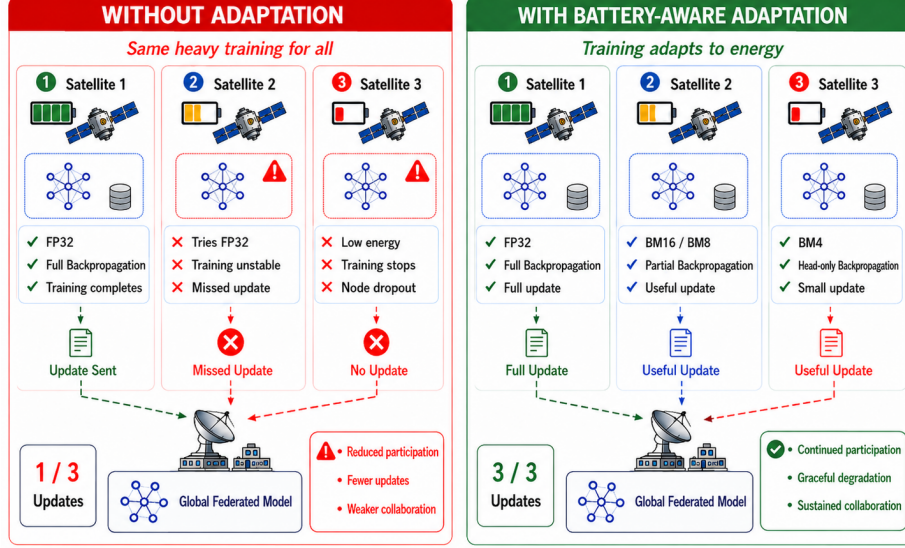


Fig. 1: A Motivating Example

2 Related Work

Service computing is widely applied across diverse areas of computing, enabling flexible integration and delivery of services [12, 23, 32]. Trust is a key component of service computing [24–26]. It supports reliable service composition and collaboration in distributed paradigms such as federated learning [10]. Federated learning has recently emerged as a service-composition mechanism for delivering distributed intelligence across low Earth orbit (LEO) satellite constellations. Elmahallawy and Luo proposed FedLEO, which improves satellite federated learning by introducing intra-plane model propagation and sink satellite scheduling to reduce the communication burden between satellites and ground stations [7]. Lin et al. proposed FedSN, a federated learning framework for heterogeneous LEO satellite networks that addresses differences in computation, memory, communication capability, and model staleness through sub-structure training and pseudo-synchronous aggregation [15]. These works mainly focus on communication topology, aggregation timing, and satellite heterogeneity, whereas our work focuses on adapting the onboard training service itself according to available energy on the satellite.

Energy-aware federated learning has also received increasing attention as an adaptive service composition mechanism to sustain distributed training services in satellite and edge-assisted learning environments. Razmi et al. proposed an energy-aware computation time scheduler for satellite federated learning that accounts for solar harvesting, eclipse periods, and battery usage, demonstrating that careful scheduling can extend battery lifetime without degrading convergence speed [20]. More recently, Yang et al. proposed CroSatFL, an energy-efficient hierarchical federated learning framework for satellite edge computing that reduces communication energy through on-orbit clustering, straggler handling, and cross-aggregation [29]. Although these studies treat energy as an important scheduling or communication constraint, they do not directly investigate how local training precision and CNN backpropagation depth should be adapted according to available energy.

Low- and mixed-precision training services aim to reduce the computational and memory cost of neural network training. Fox et al. introduced block minifloat representations for end-to-end deep neural network training, showing that 4–8 bit weight, activation, and gradient tensors can provide efficient arithmetic while maintaining competitive model accuracy [8]. Guo et al. investigated single-batch CNN training using block minifloats on FPGAs, demonstrating the relevance of block minifloat arithmetic for hardware-accelerated CNN training [9]. Li et al. proposed layered mixed-precision training, where different layers can use different numerical precisions according to their contribution to training performance, improving training speed while controlling accuracy loss [13]. These works establish the value of low-precision computation, but generally select a fixed precision based on model or hardware characteristics rather than dynamically adapting precision as a satellite service response to changing battery conditions.

Several studies have explored backpropagation as a service-level control mechanism for reducing the cost of on-device training. By limiting how much of the network participates in gradient computation, resource-constrained devices can lower the execution cost of the training service while maintaining partial learning capability. Cai et al. proposed TinyTL, which reduces memory consumption for on-device transfer learning by freezing most network weights and updating only lightweight bias modules, thereby avoiding the need to store many intermediate activations [4]. Lin et al. introduced Tiny Training Engine, an algorithmic co-design framework for on-device training under 256KB memory that uses quantization-aware scaling and sparse updates to skip gradient computation for less important layers and sub-tensors [14]. R'ub et al. proposed TinyProp, an adaptive sparse backpropagation method that dynamically adjusts the backpropagation ratio during TinyML on-device learning to reduce computation while limiting accuracy loss [21]. These approaches demonstrate that reducing the depth or scope of backpropagation can make training feasible on constrained devices; however, they do not consider satellite-specific energy dynamics or jointly combine depth control with adaptive precision in a federated learning setting.

3 Proposed Framework

This section introduces our novel Run-time Adaptive Mixed-Precision (RAMP) framework to sustain training service in satellite constellations under reduced energy conditions, i.e., solar eclipses. In RAMP, each satellite performs local training according to its current energy state, computational capacity, and assigned training configuration. Figure 2 presents steps in RAMP.

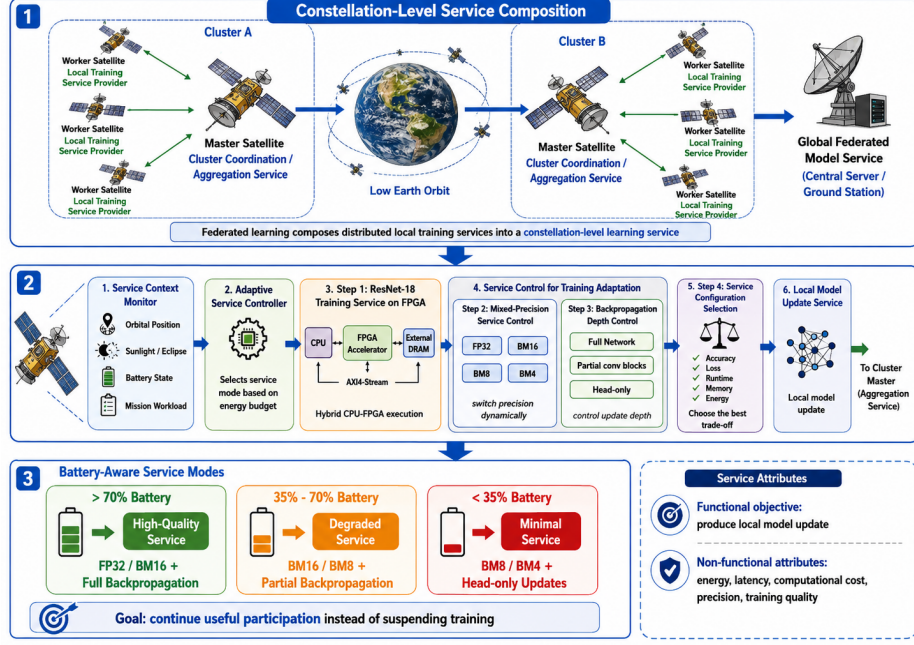


Fig. 2: Run-time Adaptive Mixed-Precision Training Framework

3.1 Resource-Constrained FPGA Realization of ResNet-18 Training

The first stage of the proposed framework implements ResNet-18 as the core onboard training service within each worker satellite. Unlike a conventional implementation of ResNet-18, the objective here is to realize a trainable CNN service within the constrained resource envelope of an FPGA-based satellite platform. This is challenging because training requires not only forward inference, but also storage and movement of intermediate activations, gradients, batch-normalization statistics, optimizer states, and model parameters. Therefore, fitting the complete training workflow on a ZCU102-class FPGA platform requires careful organization of computation, memory, and data movement.

The ResNet-18 training service is implemented in High-Level Synthesis (HLS) and adapted for small-image classification workloads with $3 \times 32 \times 32$ inputs. The network follows the standard residual structure, consisting of an initial convolutional stem followed by four residual stages with increasing channel widths.

Residual connections are explicitly supported, including projection shortcuts for stages where spatial resolution or channel dimensions change. To make the network suitable for FPGA execution, convolutional layers are implemented as reusable compute kernels, while intermediate feature maps are stored in structured on-chip buffers whenever possible and spilled to external memory when their size exceeds local storage capacity. The rationale behind creating this implementation on an FPGA is not merely a hardware acceleration exercise; the FPGA offers the flexibility to run these adaptive service modes on actual hardware and study energy-aware service adaptation in satellite federated learning. By structuring ResNet-18 training as a reusable FPGA-based service, the framework can later adjust its non-functional attributes, such as precision level, memory traffic, execution cost, and backpropagation depth, according to the available battery budget. This design enables the same local training service to operate in high-quality, degraded, or minimal modes depending on the satellite’s operational condition.

3.2 Energy-Adaptive Mixed-Precision Service Control

Building on the FPGA-feasible ResNet-18 training service introduced in the previous stage, the second stage equips the service with run-time adaptive precision control. As represented by the mixed-precision control component in Fig. 2, numerical precision is treated as a configurable non-functional attribute of the onboard training service rather than a fixed hardware choice. The functional objective remains unchanged, i.e., producing a useful local model update while the service dynamically adjusts its arithmetic cost, memory, latency, and energy consumption according to the satellite’s available energy budget.

The proposed design supports FP32 and custom block-mantissa formats, including BM16 and BM8. In the block-mantissa representation, each tensor is partitioned into blocks of 32 values, with all values in a block sharing a common exponent. BM16 represents each value using one sign bit and a 15-bit mantissa, whereas BM8 uses one sign bit and a 7-bit mantissa. This shared-exponent structure reduces arithmetic complexity and data movement while retaining a wider effective dynamic range than conventional fixed-point representations. Importantly, quantization is applied throughout the training service rather than being restricted to weights or activations. Inputs, model parameters, intermediate activations, logits, gradients, optimizer states, learning-rate values, and momentum coefficients are represented according to the precision mode active at training step t , denoted by P_t . For a block of values $\mathbf{x} = x_1, \dots, x_n$, the shared exponent is determined by the largest absolute value in the block:

$$e = \left\lfloor \log_2 \left(\max_i |x_i| \right) \right\rfloor, \quad (1)$$

and each value is projected onto the corresponding block-mantissa grid as

$$\hat{x}_i = \text{round!} \left(\frac{x_i}{2^e} 2^{m-1} \right) 2^{e-(m-1)}, \quad (2)$$

where m denotes the mantissa width, with $m = 15$ for BM16 and $m = 7$ for BM8. The reconstructed value $\hat{x}_i$ is subsequently used during forward propagation, backward propagation, and parameter updating.

Switching precision during an active training process is substantially more challenging than training a model in a single low-precision format from initialization. An abrupt transition from FP32 or BM16 to BM8 simultaneously perturbs model parameters, activations, gradients, and momentum states. In particular, optimizer states accumulated in the higher-precision domain may no longer be aligned with the representable values of the lower-precision format. The shared exponent selected for each block can also alter the effective resolution of small values. These effects can produce a sudden increase in quantization error, causing a transient loss spike, unstable gradients, or, in severe cases, failure to recover the previous convergence trajectory. To mitigate this disruption, the proposed service controller employs a precision-switching window around the scheduled transition point. During the pre-switch portion of this window, parameters and momentum states are projected onto the target precision grid, allowing the training state to become aligned with the numerical range of the lower-precision service mode before it becomes fully active. At the transition point, the forward, backward, and update paths switch to the target format. A short post-switch warm-up is then applied in which the learning rate and momentum are temporarily moderated and gradually restored.

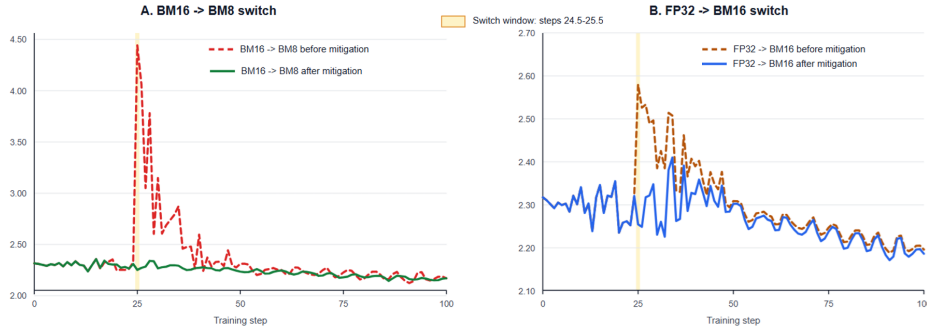


Fig. 3: Precision Switching During ResNet18 Training

This staged transition avoids exposing the model to a single abrupt numerical discontinuity and reduces the loss spike commonly observed during direct precision switching are shown in Figure 3. The onboard service minimizes the softmax cross-entropy loss over a mini-batch of N samples:

$$\mathcal{L}(\theta_t) = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{i,k} \log \hat{y}_{i,k}, \quad (3)$$

where K is the number of classes, $y_{i,k}$ is the ground-truth label, and $\hat{y}_{i,k}$ is the predicted probability produced under the active precision mode P_t . Gradients and optimizer states are quantized consistently with this mode:

$$\begin{aligned}
g_t &= Q_{P_t}(\nabla_{\theta} \mathcal{L}(\theta_t)), \\
v_t &= Q_{P_t}(\mu_t v_{t-1} + g_t), \\
\theta_{t+1} &= Q_{P_t}(\theta_t - \eta_t v_t).
\end{aligned} \tag{4}$$

where Q_{P_t} denotes quantization under the current service precision, η_t is the learning rate, and μ_t is the momentum coefficient. Both parameters are controlled during the switching window to stabilize the transition. Consequently, mixed precision functions as a service-level adaptation mechanism.

Service-Level Adaptation of Backpropagation Scope This stage adapts the scope of backpropagation according to the satellite’s energy condition. This component corresponds to the backpropagation adaptation block in Fig. 2. The complete ResNet-18 forward pass is retained to produce predictions, while the local training service selectively restricts gradient computation and parameter updates to later parts of the network. This reduces computation, activation storage, and memory traffic without suspending local training entirely.

A conventional ResNet-18 consists of an initial convolutional stem, four residual stages, and a final classification head. The proposed service exposes these components as progressively smaller update scopes. Under sufficient energy, the high-quality service mode updates the complete network. Under moderate constraints, the degraded service mode freezes the stem and earlier residual stages and updates only the final residual stage and classification head. Under severe constraints, the minimal service mode freezes the feature-extraction backbone and updates only the classification head. Consequently, the functional objective of producing a local model update remains unchanged, while the non-functional cost of the service is reduced. The update scope is selected using both the available battery budget and the expected duration of the constrained period. A short reduction in energy may be handled through precision adaptation while retaining a relatively deep backward path. However, during longer eclipse or low-energy intervals, repeated full-network backpropagation would continue to impose substantial computation and memory costs. The service therefore adopts a shallower update scope, such as the final residual stage and head, or head-only training when the constrained period is expected to persist.

Eclipse-Aware Empirical Profiling of Training Configurations The final stage of the framework, shown as Step 4 in Fig. 2, combines the adaptive mechanisms introduced in the preceding stages. The mixed-precision controller provides the available arithmetic modes, while the backpropagation controller determines the scope of the network to be updated. These choices are then combined into candidate operating configurations for the FPGA-based local training service. Rather framing it as an optimization problem, we systematically construct and empirically evaluate a finite catalogue of configurations under representative battery and eclipse conditions. Each candidate configuration is represented as

$$C_i = (P_i, S_i, \eta_i, \mu_i), \tag{5}$$

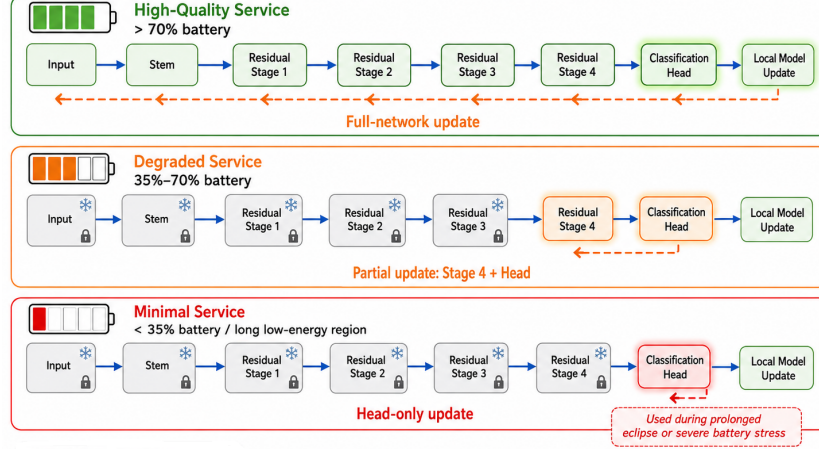


Fig. 4: Service-Level Adaptation of Backpropagation scope in ResNet-18

where P_i denotes the numerical precision, S_i denotes the selected backpropagation scope, η_i is the learning rate, and μ_i is the momentum. The evaluated catalogue includes FP32 with full-network updates, BM16 with full or late-stage updates, BM8 with late-stage or head-only updates, and a suspend-training baseline. For each configuration, we empirically record its validation accuracy, training-loss behavior, runtime, memory demand, and estimated energy cost.

The battery state evolves according to the energy consumed by the active training configuration and the energy harvested from the environment:

$$B_{t+1} = B_t - E_{\text{train}}(C_t) + E_{\text{harvest}}(t), \quad (6)$$

where $E_{\text{train}}(C_t)$ is the estimated cost of executing configuration C_t , and $E_{\text{harvest}}(t)$ depends on solar availability. During eclipse, the harvested energy decreases substantially, making the battery level at eclipse entry and the expected eclipse duration important configuration constraints. The empirical comparison is therefore performed separately for different eclipse scenarios. Among the configurations that satisfy the operational constraints, the framework identifies the least-degraded service mode that maintains stable learning and continued participation in federated learning. A short eclipse entered with a relatively high battery level may support higher precision and a broader backpropagation scope. In contrast, a prolonged eclipse or a severely constrained battery may require BM8 training with late-stage or head-only updates. Consequently, no single configuration dominates across all operating conditions; the selected configuration depends on the duration and severity of the eclipse, the available battery reserve.

4 Experimentation and Results

Baseline 1: Fixed-Precision Full-Network Training To establish a fixed-precision baseline, we compare the loss-convergence behavior of ResNet-18 on MNIST (a dataset of 70,000 grayscale digital images), CIFAR-10, and EuroSAT

under three training configurations: FP32, BM16, and BM8, as shown in Figure 5. The x-axis reports mini-batch training steps, while the y-axis shows the training loss. FP32 provides the fastest and most stable convergence across all datasets, reaching the lowest final loss because weights, activations, gradients, and optimizer states are preserved with full single-precision arithmetic. BM16 closely follows FP32, with only a moderate increase in final loss, indicating that 16-bit block floating-point representation retains sufficient numerical resolution for both forward and backward propagation. BM8 shows a larger convergence penalty, particularly at larger block sizes, due to stronger quantization effects shared across each block. The degradation is most visible on CIFAR-10 and EuroSAT, where the visual classes are more complex than MNIST and therefore more sensitive to reduced gradient and activation precision. Overall, the results show that BM16 provides a practical trade-off between convergence quality and reduced arithmetic cost for onboard training, while BM8 requires smaller block sizes to approach FP32 behavior.

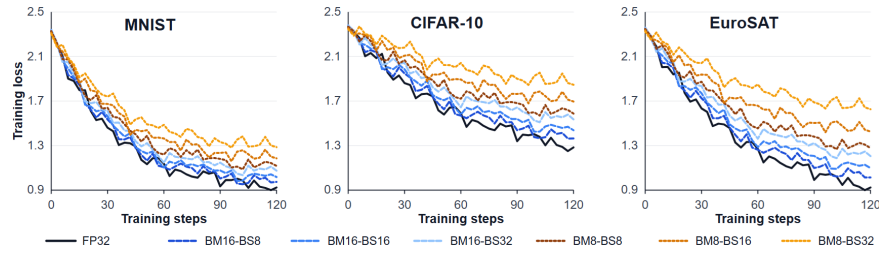


Fig. 5: ResNet-18 Loss Convergence Across Precision Formats

Baseline 2: Fixed-Precision Training with Skipped Steps This experiment establishes a non-adaptive baseline for energy-constrained satellite training. The onboard ResNet-18 model is trained using FP32 precision, while selected mini-batch update steps are intentionally skipped (randomly) to emulate eclipse or low-energy operating windows during which local optimization cannot be performed. In these intervals, the satellite does not execute the training process, and therefore contributes no local model improvements to the federated learning process. This setting represents a non-adaptive baseline, where training is suspended rather than compensated through precision scaling. The objective is to quantify how missed updates affect loss convergence over time. Figure 6 shows the resulting FP32 loss trajectories under uninterrupted training and skipped-update training, with the low-energy windows explicitly marked.

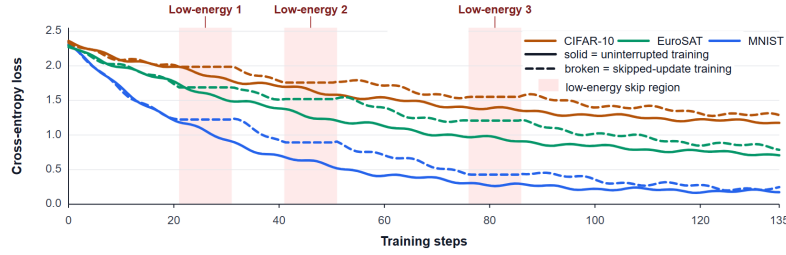


Fig. 6: FP32 ResNet-18 Loss Response to Skipped Training Updates

Mixed-Precision Training This experiment evaluates whether local training can continue during eclipse periods by degrading numerical precision instead of skipping training steps. We use the same energy schedule as the skipped-step experiment, but replace missed updates with reduced-cost training updates: the model runs in FP32 during nominal-energy intervals and switches to lower-cost block-mantissa formats, such as BM16, during low-energy intervals. Figure 3 shows that the loss converges if we switch to lower precisions during training.

Depth Control in Backpropagation This experiment evaluates the impact of controlling backpropagation depth during eclipse and low-battery intervals. In this configuration, forward propagation is maintained for every training step, while gradient computation and parameter updates are selectively limited to the later stages of ResNet-18, such as the final residual blocks and the classification head. This hierarchical update policy reduces the computational and memory burden of local training while still allowing the model to adapt to incoming data. The experiment compares full backpropagation, precision-aware depth-controlled training, and a skipped-step baseline in which training is suspended during low-energy periods. As shown in Figure 7, reducing numerical precision leads to a corresponding decrease in accuracy; however, the depth-controlled configurations continue to make useful training progress under constrained energy conditions. In comparison with the suspended-training behavior shown in Figure 3, hierarchical backpropagation achieves higher accuracy.



Fig. 7: Limited Back Propagation During Training

Layer-Wise Mixed Precision in ResNet-18 This experiment investigates mixed-precision training within ResNet-18 network. Instead of assigning a sin-

gle precision to the entire network, the proposed scheme progressively reduces precision in later network layers while keeping earlier layers at higher precision. This design is motivated by the fact that different parts of a deep network can exhibit different sensitivities to quantization. The objective of this experiment is to determine whether layer-wise mixed precision can reduce computational cost while preserving convergence behavior and accuracy. We compare this approach against suspending training once a resource or precision threshold is reached. As shown in Figure 8, the suspended-training baseline initially follows the same warm-up trajectory, but then plateaus once updates are halted. Whereas, the layer-wise mixed-precision strategy continues training after later layers are moved to lower precision and achieves a stronger accuracy trajectory.

Combined Mixed Precision and Backpropagation Scope This experiment evaluates the complete adaptive training service by jointly combining mixed-precision execution with depth-controlled backpropagation. Under nominal energy conditions, the satellite selects higher-capacity training configurations that use higher numerical precision and propagate gradients through a larger portion of the model. During eclipse and low-energy intervals, the service transitions to lower-cost configurations by reducing both the arithmetic precision and the backward update depth. Figure 9 illustrates the resulting behavior of mixed-precision training with hierarchical backpropagation depth control.



Fig. 8: Mixed Precision in Model's Layers During Training



Fig. 9: Mixed Precision Combined with Back Propagation Depth Control During Training

5 Conclusion

This paper presents an energy-aware federated learning framework that treats onboard AI training as an adaptive service for energy-constrained satellite constellations. The framework combines mixed-precision quantization using FP32, BM16, and BM8 with hierarchical depth-controlled backpropagation, enabling satellites to continue producing degraded but useful local updates instead of suspending training during low-energy periods. Using ResNet-18 and evaluations on CIFAR-10, MNIST, and EuroSAT, the results across six experimental settings show that reduced-precision and reduced-depth training can preserve convergence more effectively than skipping training steps. In particular, the combined precision- and depth-adaptive strategy provides the strongest balance between model performance, computational cost, and continued service participation under dynamic orbital energy conditions. The design is developed through an HLS-based implementation considering the resource capabilities of the Xilinx ZCU102

FPGA platform. Future work may include hardware validation on FPGA platforms, precision- and depth-aware aggregation at the global model level.

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